Vision and Need

Although the methods behind motion capture are well documented, researchers must often rebuild the infrastructure to assemble them into a usable measurement system when existing tools do not fit their needs. Commercial marker-based motion capture systems offer sophisticated general-purpose measurement, but remain costly and closed to modification. Newer markerless systems, driven by advances in computer vision, have lowered those barriers but are often built around the needs and assumptions of particular fields. Researchers working outside those assumptions (for instance, anyone measuring a non-human animal) are left to learn and implement components common to multi-camera markerless motion capture systems themselves. These include video recording and synchronization, camera calibration to recover each camera's position and optical parameters, 2D pose estimation using AI models to locate anatomical keypoints, 3D reconstruction by triangulating those keypoints across camera views, and post-processing and analysis. Rebuilding this infrastructure across projects duplicates technical effort and reduces the return on public research investment in software that is difficult to maintain, reproduce, or transfer beyond the environment in which it was developed.

There is therefore a need for accessible, shared research infrastructure that allows specialized motion capture adaptations developed by one community to be contributed back and reused by others, requiring defined stewardship and governance as much as software.

FreeMoCap [16] provides an existing technical foundation for this model. Maintained by the FreeMoCap Foundation, FreeMoCap is free and open-source software that transforms synchronized video from consumer-grade cameras into research-grade 3D motion-capture data through a modular workflow (Figure 1). This proposal requests Track 1 support to **scope the organizational and infrastructure requirements needed to transition FreeMoCap from an open-source project maintained by a small core team into a secure, sustainable, community-driven ecosystem.**

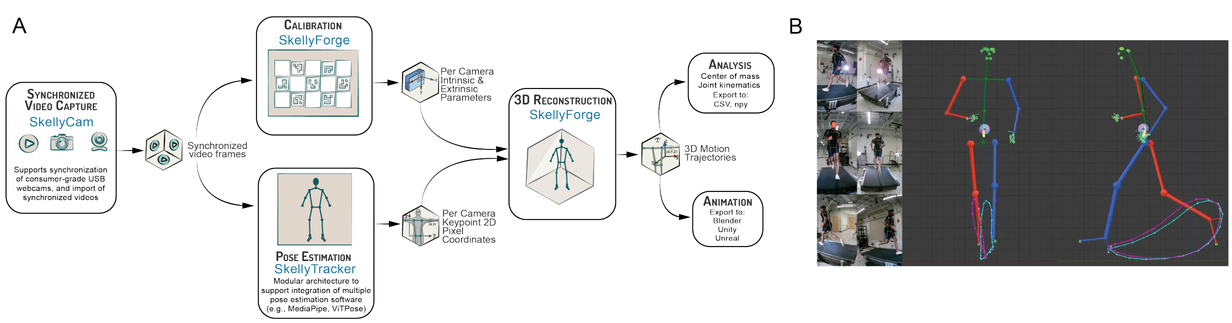

Figure 1: The FreeMoCap workflow and its component repositories (blue). (A) Synchronized video from consumer-grade USB webcams (skellycam) is processed through camera calibration and 3D reconstruction (skellyforge) and modular pose estimation (skellytracker), producing 3D motion trajectories for downstream analysis or animation export. Separation of the pipeline by domain allows individual components to be modified or replaced independently. (B) Reconstructed 3D trajectories alongside the source camera views.

Current State of the Software and Community

FreeMoCap, distributed through Github and a standalone desktop installer (freemocap-dot-org-slash-download), has over 10,000 Github stars, telemetry indicating over 15,000 users across 153 countries, and a community forum with over 4,000 members. The software is organized as interoperating repositories separated by domain: `skellycam` for camera synchronization and recording, `skellytracker` as a modular interface for pose estimation, `skellyforge` for calibration, 3D reconstruction, and post-processing, and `freemocap_blender_addon` for export to Blender.

FreeMoCap is used across several distinct communities, including researchers, educators and students, creative and technical users, and software contributors. Published use of FreeMoCap spans reaching and grasping biomechanics [9], gait analysis [20], computational linguistics [13, 15, 12], and animation [7]. Independent validation studies have reported agreement with marker-based motion capture across several movement tasks [5, 10], while our own work [2] has characterized gait and balance performance, demonstrated integration of multiple pose-estimation backends, and extended the workflow to non-human locomotion (Figure 2).

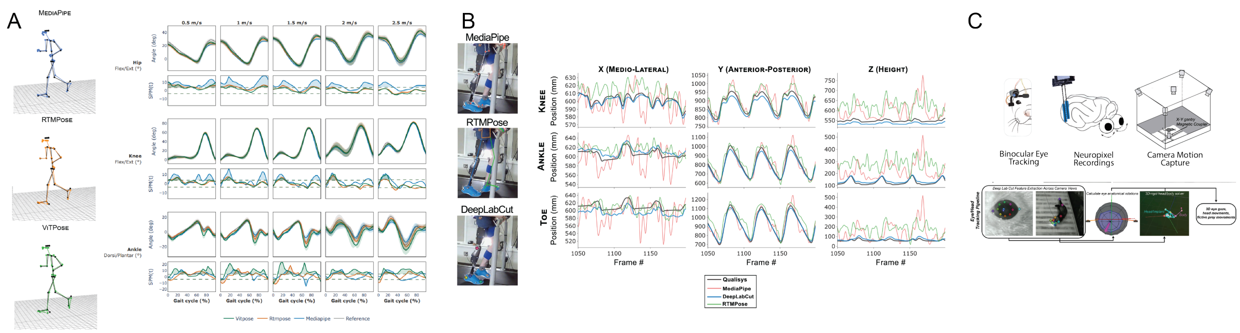

Figure 2: FreeMoCap applications from standard human biomechanics to specialized movement-analysis use cases. A. Lower-extremity joint-angle comparison and statistical parametric mapping against Qualisys marker-based motion capture across multiple pose-estimation backends. B. Custom DeepLabCut integration for tracking a lower-limb prosthesis user. C. Extension to ferret locomotion and neural-behavioral research in collaboration with the Scholl Lab, CU Denver Anschutz

While user adoption has grown, it has outpaced the growth of the maintainer base. Much of the architectural and operational knowledge needed to contribute to the core pipeline remains concentrated within a core development team, while contributor onboarding and technical knowledge decentralization remains immature.

Team Qualifications

The FreeMoCap team combines scientific and engineering expertise with more than five years of direct experience developing the FreeMoCap software, supporting its community, and managing the technical and organizational challenges associated with its growth.

Ecosystem Discovery

Need for Innovation

The available tools demonstrate substantial demand for accessible markerless motion capture, but each is built around particular assumptions about users, domains, hardware, or deployment. Commercial systems include biomechanics-focused platforms such as Theia3D and animation/production tools such as Move.ai and Rokoko, while open-source alternatives such as OpenCap [18], Pose2Sim [14], and PosePipe [4] are largely centered on human biomechanics. These systems also impose different operational constraints: Theia3D relies on expensive hardware, while OpenCap's standard workflow depends on supported iOS devices and cloud processing. The continued development of domain-specific workflows in individual research laboratories [8, 19, 11, 3] indicates that researchers with specialized requirements still build substantial portions of this infrastructure independently.

Our Track 1 scoping will test the premise underlying this proposal: **repeated rebuilding of motion-capture pipelines is an ecosystem failure, as starting over is often easier than extending shared infrastructure.** Track 1 will determine what documentation, contribution pathways, technical interfaces, and governance structures are needed to reduce barriers to extending and contributing to shared research infrastructure.

Necessity of an OSE approach

FreeMoCap's existing community provides early evidence that an OSE can convert individual expertise into shared infrastructure for the broader user community. Users have already become contributors in areas including animation, UI design, and pose-estimation architecture, bringing domain expertise from outside the core development team into the project, for the benefit of all users. However, these transitions occurred informally and have depended heavily on direct interaction with the core team. Creating a repeatable path into the ecosystem first requires understanding who is trying to use, teach, extend, or contribute to FreeMoCap, what technical knowledge each group needs, and where the current software and documentation fail to support them independently.

Stakeholder discovery

Although telemetry indicates more than 15,000 users, our current understanding of the FreeMoCap community is drawn primarily from users who contact us directly or participate in existing community channels. Some applications have emerged outside the domains we originally anticipated. For example, computational-linguistics researchers have adopted FreeMoCap in published work [13, 15, 12] without any prior interaction with the core team. This motivates a broader discovery effort aimed not only at known users, but also at understanding unexpected communities, what drew them to the platform, and what would reduce barriers to sustained and broader adoption.

Our proposed work will combine analysis of existing community activity, community-wide surveys, and the I-Corps for PESOSE stakeholder-discovery process to characterize four user populations reflecting the communities already observed around FreeMoCap: (1) researchers across application domains, (2) educators and students, (3) creative and technical users in animation and related fields, and (4) current and prospective contributors. We will also interview maintainers of mature open-source projects, not as FreeMoCap users, but as a source of comparative evidence on how successful ecosystems structure contribution, governance, and extension. Discovery will produce three assessments, each addressing one condition under which extending shared infrastructure becomes easier than rebuilding it.

1) **Cross-domain needs assessment**: how different communities use FreeMoCap, what brought them to it, and what their work requires that the shared pipeline does not yet provide.

2) **Knowledge assessment**: what users must understand to operate the software, what contributors must understand to modify it, which of that knowledge remains undocumented, and what forms of documentation or instruction would allow people to participate without direct access to the core team.

3) **Comparative ecosystem review**: how mature open-source projects of comparable scale structure contribution, extension, and decision-making, and how those structures perform as projects grow. These findings will supply candidate models for evaluation in Organization and Governance and inform the security, community-building, and sustainability work that follows.

Organization and Governance

Working in Public [6], an analysis of open-source maintenance and sustainability, distinguishes open-source projects by the growth of their user and contributor bases. FreeMoCap currently resembles a *stadium*: high user growth with comparatively low contributor growth, leaving a large community dependent on a small maintainer base. **Track 1 will build on that foundation by scoping the transition toward a *federation*, in which user and contributor bases grow together and technical knowledge, responsibility, and authority are distributed across the ecosystem.**

Governance Scoping

Federated projects require the distribution of decision-making and knowledge such that it does not depend on any one person. Python's PEP process is one example, in which an elected steering council, trusted core contributors and editors, and a formal proposal process distribute technical authority across the community. We aim to evaluate such models against FreeMoCap's structure and scale, using interviews with maintainers of mature open-source projects during Ecosystem Discovery to supply the candidate models.

Distributed infrastructure and community extensions

A federated model requires distributed technical knowledge and the ability for contributors to extend the software without the core team implementing each addition. FreeMoCap's modular architecture provides a technical basis for this: the `skellytracker` interface provides an open interface through which AI-based pose-estimation models developed by different groups can interoperate with a shared scientific workflow without modification to the underlying pipeline. This interface matters because no single pose-estimation model is adequate across subjects. The algorithms in common use are trained on standard human datasets, and the requirements of biomechanical measurement differ from those the models were built to satisfy [17].

A community extension, in this context, is code written and maintained by someone outside the core team that runs against a published interface without altering the shared pipeline. A researcher whose subjects fall outside existing models could implement a tracker against a documented interface rather than assembling an entire pipeline around it, and the result would be available to everyone facing the same problem, while an AI method published by a computer-vision researcher can be translated into a reusable scientific workflow for users who would otherwise lack the technical expertise to implement it. Track 1 will define the review, distribution, maintenance, and responsibility structures required for this model, informed by projects already operating extension ecosystems. The resulting framework could support other community-developed capabilities, including biomechanical analyses.

Licensing

FreeMoCap and its component repositories are licensed under the AGPLv3+, on the premise that improvements to publicly funded infrastructure are most valuable when they return to the broader community. The Foundation retains the ability to offer alternative terms to organizations whose intended use is incompatible with the AGPLv3+, keeping the software freely available while generating resources for maintenance and stewardship.

Long-term Sustainability

The FreeMoCap Foundation was incorporated as an IRS-recognized 501(c)(3) public charity to provide an institutional home for the project, and the governance structures scoped above are intended to give that home a distributed contributor base rather than a dependence on any particular maintainer. In parallel, the Foundation will evaluate a diversified model for the long-term sustainability of this research infrastructure, including alternative commercial licensing, standardized hardware kits, training and support services, institutional partnerships, and continued grant funding. SkellyTech, LLC, a separate for-profit entity associated with the project, provides an additional pathway for commercial services and other revenue-generating activities that may support the long-term sustainability of the ecosystem.

Risk Analysis and Security Plan

Video-based motion capture creates inherent privacy concerns because identifiable recordings must be captured and processed. These concerns can themselves become a barrier to adoption: recent work on OpenCap found that patients remained concerned about video sharing, storage, and access despite HIPAA-compliant cloud safeguards [1]. FreeMoCap avoids this cloud-data surface by processing video and derived data locally, without requiring a network connection, allowing sensitive recordings to remain under the user's direct control.

Secure and trustworthy research infrastructure also requires addressing software-supply-chain risk. FreeMoCap depends on a substantial stack of computer-vision, machine-learning, and scientific-computing libraries, and the proposed community extension system would introduce additional third-party code into the processing environment. Track 1 will assess FreeMoCap against security standards maintained by the Open Source Security Foundation (OpenSSF), which provides established criteria and certification pathways for secure open-source development. This assessment will identify gaps and the requirements needed to reach an appropriate certification level, informing both a project security roadmap and the review, approval, and release controls needed to guard against malicious or inadvertently unsafe community extensions.

Data Integrity

While the fidelity of FreeMoCap-produced data has been and continues to be validated, **preserving the integrity of that data (i.e., that modifications to the software do not silently alter the measurements it produces) is an ecosystem responsibility**. Track 1 will scope stage-specific integrity checks across the processing pipeline, with expected outputs tracked across software versions. These may include calibration checks against known board geometry, 2D pose-estimation consistency, 3D reconstruction metrics such as positional differences, smoothness, and jerk, comparisons against reference datasets, and checks before and after each post-processing operation so that the effect of filtering or other transformations is explicitly measured. Camera characteristics are also part of the measurement chain. Collaboration with Imatest LLC, whose work includes ISO/TC 42 standards activity on image-information content, will inform how imaging-system limitations can be characterized, reported, and made visible to users.

Identity and Telemetry

Potential ecosystem features such as contributor recognition, authenticated accounts, and usage telemetry introduce additional privacy and security considerations. Track 1 will scope approaches that minimize collected data and preserve FreeMoCap's local-first principles, including evaluation of established authorization approaches such as OAuth 2.0 for account-based features. The project's industry mentor, Endurance Idehen, will advise this work, including authentication, authorization, telemetry, and related security controls.

Together, these activities will produce a risk register, draft extension review and release controls, and a data-integrity and security roadmap.

Community Building

Community building is a critical part of the transition from a *stadium* to a *federation* ecosystem. **A larger contributor base only reduces dependence on the core team if users and contributors have enough knowledge to act independently; otherwise, growth simply shifts effort toward support, review, and correction.**

The FreeMoCap Foundation maintains an active Discord community forum of more than 4,000 members, structured so that users can ask and answer questions without relying exclusively on the core development team. The core team also hosts a weekly community call for project updates, user feedback, technical discussion, and interaction with potential contributors. These activities provide an established foundation for community engagement, but rapid user growth in 2026 (Figure 3) **has increased the need for education and participation models that can scale beyond direct interaction with maintainers.**

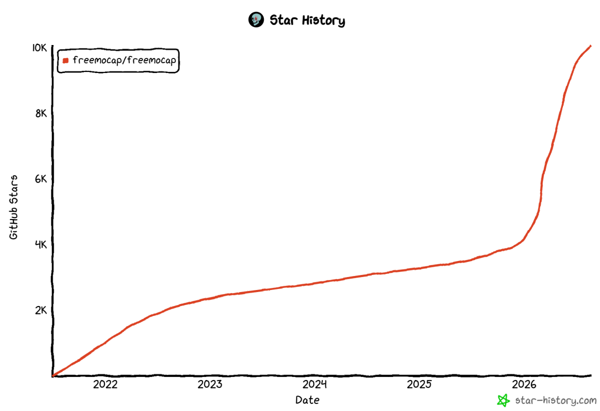

Figure 3: Github star chart

Track 1 will develop and pilot accessible educational and participation pathways at three levels: **introductory materials** for general users; **intermediate and domain-specific resources** for scientific, technical, creative, and educational applications; and separate **developer documentation** covering software architecture, contribution workflows, testing, and maintenance responsibilities.

These materials will be piloted through hands-on classes and workshops with partner organizations, including The Possible Zone and Artisans Asylum, and through tutorials and guided projects made available to the broader community. Pilots will identify where learners encounter conceptual or technical barriers, which formats work best for different audiences, and what knowledge can be transferred without direct core-team instruction.

Track 1 will also test low-barrier forms of community participation that do not require software development, including community-created visualizations, artwork, examples, challenges, and other ways for users to contribute domain expertise. Together, these activities will inform a scalable education and onboarding model that supports progression from user, to informed participant, to sustained contributor. Longer term, the resulting model could inform a larger FreeMoCap community meeting or conference bringing together users, developers, educators, and researchers as part of a subsequent Track 2 effort.

Milestones and Evaluation Plan

Qtr	Milestone / Deliverable	Evaluation
Q1	Complete I-Corps for PESOSE; establish baseline ecosystem metrics	Completion confirmed by I-Corps instructors
Q1–Q2	Stakeholder discovery; prioritized user and contributor barrier map	Target groups covered; recurring barriers confirmed
Q2	Develop candidate governance, contributor, and stewardship models	Compared with discovery findings and reviewed by OSE maintainers
Q2–Q3	Define extension contribution, maintenance, licensing, and review requirements	Tested against discovered extension needs; external OSE review
Q3	Security/data-quality scoping; prototype benchmarking and regression suite	Known output differences detected; risks mapped to controls
Q3	Pilot education and onboarding with partner organizations	Participant feedback and observed barriers
Q4	Integrate findings into governance, sustainability, community, and implementation roadmap	Processes usable without direct core-team consultation

Evaluation

Track 1 success will be evaluated by: (1) documented governance roles, contributor pathways, and software architecture, assessed by whether contributors outside the core team can act on them without direct consultation; (2) educational and onboarding materials, assessed through pilot participant performance and observed barriers; and (3) a framework for developing, reviewing, distributing, and maintaining community extensions, assessed through limited pilots of the proposed extension process. Each output will be revised in response to stakeholder feedback before incorporation into the final ecosystem plan.

Broader Impacts

In a fully developed ecosystem, the same software can serve a thirteen-year-old with a webcam and a research laboratory measuring clinical populations. A contribution made by an animator can improve the software for researchers and clinicians, and scientific advancements can become accessible to artists, educators, students and hobbyists. A sustainable FreeMoCap ecosystem can turn specialized adaptations into shared infrastructure, increasing the return on public investment in scientific software while creating pathways for expertise to move across fields.

The educational activities piloted under this proposal will help develop and test materials for users of differing technical backgrounds to progress towards independent software use, and where desired, contribution. A mature ecosystem can help a user understand how a pixel on an image can become a 3D point in space with the software, rather than just operating it. By pairing extensible infrastructure with education and cross-domain participation, FreeMoCap, supported by a developed ecosystem, can broaden not only who has access to quantitative measurement tools, but also who is able to understand, adapt, and help shape them.